PROJECT DOCUMENT

Mitsue Project

Sugano Organic — Biomass CHP Pilot · Technical Requirements & Proposal

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Author	Rob Oudendijk

Sugano Organic — Biomass CHP Pilot

Technical Requirements & Proposal · Phase 1 Functional Prototype

Prepared for: Tokuo Aomi (青海 徳生), Sugano Organic
Prepared by: Rob Oudendijk, Mitsue Project
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Purpose

This document outlines the technical requirements and investment case for installing a small biomass combined heat and power (CHP) system at Sugano Organic, powered entirely by Sugi trees from Tokuo's own mountain forest.

Sugano Organic becomes **Phase 1 of the Mitsue Project's energy programme** — a real, working prototype demonstrating that a rural home and workshop can be fully powered and heated by the forest around it. Every product made at Sugano Organic thereafter carries that proof.

Site Overview

Item	Detail
Location	Mitsue Village, Nara Prefecture
Electricity need	~10 kW peak (house + workshop)
Heating need	~23 kW (winter, house + factory)
Fuel source	Tokuo's own 28.05 ha Sugi mountain forest
Grid connection	Retain as backup / export surplus

System Design

Core: 10 kW Wood Gasifier + Generator

A wood gasifier converts Sugi logs and chips into combustible gas (syngas), which drives a generator producing electricity. Waste heat from the engine is captured for space and water heating.

Parameter	Value
Electrical output	**10 kW**
Thermal input required	45.5 kW
System efficiency (electrical)	22%
Wood consumption	**10.9 kg/hour**
Daily consumption (8h operation)	87 kg/day
Annual consumption (250 days)	**21.8 tonnes/year**

Parameter	Value
Annual electricity generated	**20,000 kWh/year**

Heat Recovery (CHP)

The engine produces far more heat than electricity. Capturing this waste heat raises overall system efficiency from 22% to 84%.

Parameter	Value
Waste heat available	35.5 kW
Recoverable (80% efficiency)	**28.4 kW**
House + workshop heating need	~23 kW ✓
Heating saving vs kerosene	**¥340,000/year**
Overall CHP efficiency	**84%**

CHP Components

- Water-cooled heat exchanger jacket on engine
- 500–1,000 L insulated hot water storage tank
- Piping and insulation (house + workshop circuit)
- Radiators or underfloor heating connections
- Control and integration unit

Fuel Supply: Tokuo's Forest

Parameter	Value
Forest area owned	28.05 ha
Rotation cycle	30 years
Annual sustainable harvest	0.94 ha/year → **5.6 tonnes/year**
Initial thinning (one-time)	~**196 tonnes** (35% of overgrown stock)
Fuel from initial thinning alone	**~9 years** at 10 kW consumption

Key point: Mitsue's Sugi plantations are heavily overgrown and actively need thinning for forest health. The initial clearance is not a cost — it is forest restoration work that also produces the first 9 years of fuel for free.

Long-term (after year 9), a supplementary ~16 tonnes/year can be sourced from Mitsue Village community forest — a small volume easily arranged through Mitsue Kanko or the village forest cooperative.

Wood Processing Requirements

The system requires air-dried wood chips or split logs (15–20% moisture content).

Equipment needed	Note
Chainsaw + log splitter	For initial thinning — may already be owned
Wood chipper (if chips preferred)	Rental or shared with Mitsue Kanko
Covered wood drying storage	~50 m² shed, 3–6 months drying time

Mitsue Kanko can potentially coordinate the annual thinning and processing as a service, supporting multiple village landowners over time.

Investment & Financial Summary

System Costs (Japan market, 2026)

Item	Low estimate	High estimate
10 kW gasifier + generator unit	¥2,800,000	¥5,500,000
Installation (~25% of unit cost)	¥700,000	¥1,375,000
CHP heat recovery system	¥1,500,000	¥3,500,000
Total investment	**¥5,000,000**	**¥10,375,000**
Approximate USD	$33,000	$69,000

Annual Savings

Item	Value
Grid electricity avoided	¥540,000/year
Kerosene heating avoided	¥340,000/year
Combined annual saving	**¥880,000/year**

Payback Period

Scenario	Payback
Without subsidies	5–11 years
With 50% subsidy (realistic)	**3–6 years**
CO_2 avoided	9.0 tonnes/year

Subsidy & Funding Pathways

The Mitsue Project NGO (currently being established) will actively pursue the following grants on behalf of Tokuo:

Programme	Body	Typical support
Rural biomass energy subsidy	METI / 経済産業省	Up to 50% of system cost
Agricultural biomass support	MAFF / 農林水産省	Equipment grants for wood energy
Forest thinning subsidy	Forestry Agency / 林野庁	¥50,000–¥80,000/ha thinned
Carbon credit income	J-Credit scheme	¥2,000–¥5,000/tonne CO_2

The NGO acts as the **financial and administrative tunnel** — handling applications, coordinating with government, and channelling grant funding directly to the site.

Grant interaction with the Mitsue Forest Cooperative (御杖村森林組合)

Partnering with the Forest Cooperative does **not** block our energy grants — the two funding streams are separate pots:

Pot	Who claims it	Effect on us
Forestry maintenance / thinning (森林環境譲与税, 林野庁 thinning ~¥50–80K/ha, abandoned-forest maintenance)	Cooperative / landowner, as accredited forestry operator	**Net positive** — their thinning subsidy lowers the cost of producing our fuel; we benefit indirectly, not by receiving cash
Energy / CHP capital (METI rural biomass up to 50%, MAFF agri-biomass, NEDO)	Our project / NGO / Tokuo	**Fully available** — unaffected by cooperative's forestry grants

Watch-out — J-Credit double-counting: Each tonne of CO_2 can be claimed once, by one entity. Our J-Credit angle is **fossil-fuel displacement** (kerosene + grid avoided by the CHP), a different methodology from forest-carbon *absorption* — so they can coexist if scoped separately. **Verify before assuming both:** confirm the cooperative's / landowner's current J-Credit registration status for Tokuo's parcels, and confirm the same-parcel thinning subsidy is not double-claimed in one period.

(Note: 御杖村森林組合 — also referred to locally as Mitsue Kanko — is the village's sole forestry operator, handling cutting, forest care, and the wood cutting/processing logistics named earlier in this proposal.)

Project Budget Context (EVM Baseline Rev 1)

The Sugano Organic CHP installation is a ¥5M–¥10.4M capital item — it does **not** sit inside the Phase 1 budget (¥5.5M, allocated entirely for feasibility studies and NGO setup). It aligns with the project budget as follows:

EVM Line	Phase	Budget	Role
3.3 Energy systems feasibility study	Phase 1	¥1.5M	Assess CHP viability at Sugano Organic; supplier quotes; subsidy mapping
5.x Pilot Build — CHP installation	Phase 3	TBD (within ¥177M)	Actual hardware and installation, if project-funded
METI / MAFF / Forestry Agency grants	External	Up to 50% of cost	Primary capex source — applied for by the NGO on Tokuo's behalf

Key implication: Phase 1 delivers the feasibility study and grant applications. The hardware is funded externally (subsidies) or from Phase 3 capex — it does **not** create a Phase 1 budget overrun. Tokuo's site can proceed independently of the larger project build timeline.

Prototype Value for the Mitsue Project

Tokuo's site is the ideal Phase 1 prototype because:

1. **He already owns the fuel** — no land acquisition, no supply chain to build
2. **Small, manageable scale** — 10 kW is technically simple and low-risk
3. **Dual use** — powers both a home and a creative workshop
4. **Strong story** — natural clothing + nature sounds, powered by the forest. Authentic and communicable worldwide

5. **Replicable** — dozens of Mitsue households and small businesses face the same situation

Data from this installation (fuel consumption, uptime, costs, CO_2) feeds directly into the larger Mitsue Project energy planning and grant applications.

Community Extension — EV Charging at the Former Sugano Elementary (Koryukan)

Because the Sugano CHP provides **reliable power close by**, a natural community add-on is **EV charging at the former Sugano Elementary School (Koryukan / みつえ体験交流館)** — also the leading Phase-1 candidate site for the project hub:

- **Who benefits:** parents, teachers, and visitors using the Koryukan's educational/experience programmes, plus the wider community — a visible, everyday public benefit.
- **Why it fits here:** the site already has **space and a parking area**, and now has **reliable generation nearby** (Sugano biomass CHP + complementary solar/battery). It is the natural hub for village charging infrastructure.
- **Resilience:** the CHP can keep the chargers (and the Koryukan) running during a grid outage.
- **Scope note:** this is a complementary community idea, not part of Tokuo's 10 kW home/workshop system. It would be sized and confirmed in the Phase 1 energy feasibility study and built with the project's EV charging work (Gantt WP 88 / JP 193).

Proposed Next Steps

1. **Site visit** — walk the forest together, assess thinning priorities and wood storage location
2. **Contractor shortlist** — identify 2–3 Japanese gasifier suppliers for quotes (e.g. Ankur Scientific, local fabricators)
3. **NGO grant scan** — Rob's team identifies applicable subsidy programmes and eligibility
4. **Feasibility sign-off** — confirm grid connection requirements with local utility
5. **Phase 1 decision** — agree scope, budget, and start date

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Summary

> Tokuo's own Sugi mountain — the forest his family has tended for generations — is ready to power his home and workshop completely, cleanly, and independently.
>
> A 10 kW biomass CHP system costing ¥5M–¥10M, with a realistic payback of 3–6 years after subsidies, producing 20,000 kWh of electricity and 28 kW of winter heat from wood that would otherwise go unthinned.
>
> **先人が植えた森が、今、村を支える力になる。** *The forest our ancestors planted — the power that sustains the village they built.*